



A
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on
EmoHealthNexus:Cross Domain Disease Detection
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By

Aanchal Gupta (2200290100001)
Aryan Kumar (2200290100041)
Aastha (2200290100003)
Akanksha Choudhary (2200290100015)

Under the supervision of
Dr. Preeti Garg
KIET Group of Institutions, Ghaziabad
Affiliated to
Dr. A.P.J. Abdul Kalam Technical University, Lucknow
(Formerly UPTU)
May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Name: Aanchal Gupta
Roll No: 2200290100001

Signature:

Name: Aryan Kumar
Roll No: 2200290100041

Signature:

Name: Aastha
Roll No: 2200290100003

Signature:

Name: Akanksha Chaudhary
Roll No: 2200290100015

Signature:

Date:

CERTIFICATE

This is to certify that the Project Report entitled “**EmoHealthNexus: Cross Domain Disease Detection**”, which is submitted by Aanchal Gupta, Aryan Kumar, Aastha, Akanksha Choudhary in partial fulfilment of the requirement for the award of degree B. Tech. in Department of **Computer Science & Engineering** of Dr A.P.J. Abdul Kalam Technical University, Lucknow, is a record of the candidate's own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Dr. Preeti Garg
(Associate Professor, CSE)

Dr. Vineet Sharma
Dean(CSE)

Date:

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We also do not like to miss the opportunity to acknowledge the contribution of all the faculty members of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

Date:

Signature:

Name:	Aanchal Gupta	Aryan Kumar	Aastha	Akanksha
Roll No:	2200290100001	2200290100041	2200290100003	2200290100015

ABSTRACT

Digital health technologies are transforming modern healthcare by enabling continuous monitoring and preventive medical interventions. Traditional healthcare systems mainly focus on reactive treatment after symptoms appear, while modern digital health systems aim to identify risks at an early stage. But many existing digital health systems look at either physiological or emotional data, but not both, leading to limited health insights.

EmoHealthNexus is a cross-domain **disease detection framework** approach to healthcare that uses physiological data collected from wearable devices, along with emotional and self-reported health data. It integrates real-time physiological information (heart rate, sleep, activity) from wearable devices and emotional and self-reported health information (emotional states, symptoms) via structured patient self-reports and sentiment analysis.

The system uses machine learning techniques including Random Forest Classification, Decision Tree and Logistic Regression to process data and predict early symptoms of potential risks. The data is processed using Python backend, which extracts features and runs the predictive models. The processed results are stored in MySQL database and presented through a React.js dashboard, allowing patients and doctors to track their health each week.

The system also features a **patient-doctor consultation** interface, where doctor-approved automated predictions can be made. Our experiments show that the Random Forest model outperforms Decision Tree models in predicting outcomes when using the combined dataset.

In summary, EmoHealthNexus showcases the benefits of integrating physiological and emotional health information to enhance predictive accuracy, facilitate early diagnosis and improve patient-doctor consultation in the healthcare industry.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
IOT	Internet of Things
NLP	Natural Language Processing
ML	Machine Learning
HRV	Heart Rate Variability
PPG	Photoplethysmography
API	Application Programming Interface
SQL	Structured Query Language
RF	Random Forest
DT	Decision Tree

CHAPTER1

INTRODUCTION

1.1 INTRODUCTION

The rapid advancement of digital healthcare technologies has significantly changed the way healthcare services are delivered. Wearable devices such as smartwatches, fitness bands, and smart rings allow continuous monitoring of physiological parameters including heart rate, sleep patterns, and physical activity. These devices transmit real-time health data to cloud platforms, enabling healthcare providers to monitor patients remotely and identify potential health risks earlier than traditional clinical methods.

Despite these technological developments, many existing digital health systems focus primarily on physiological measurements while ignoring emotional and psychological factors that significantly influence human health. Research studies indicate that mental conditions such as stress, anxiety, and depression can strongly affect physical health by disrupting metabolic balance and increasing cardiovascular risks.

However, most current health monitoring applications analyze physiological data and emotional data separately. This distinction constrains the capacity of health care to fully understand the state of a patient. This means that disease symptoms may be overlooked and preventive health strategies may be less effective.

To overcome this challenge, a cross-disciplinary perspective of health care is needed that considers both physical health measures and emotional states together.

1.2 PROJECT DESCRIPTION

This project introduces EmoHealthNexus, a holistic health management system that links physiological data with emotional health.

It gathers physiological data via wearable devices such as heart rate variability, sleep duration, and exercise. Meanwhile, patients also report subjective symptoms, emotions, and stress through a web portal. The information from these two sources is integrated to provide a holistic view of the patient's health.

The data is fed into a machine learning system that uses algorithms like Random Forest, Decision Tree, and Logistic Regression to examine the associations between emotional and physiological variables. It detects correlations that could indicate early onset of health problems and issues predictive warnings for preventive health care.

A dashboard developed with React.js enables users to view weekly health, activity, and emotional data patterns. It also includes a doctor consultation module, enabling doctors to access the AI-based predictions and offer professional medical feedback.

The major objectives of the EmoHealthNexus system are:

1. To combine physiological and emotional health data for monitoring health.
2. To enable early prediction of health risks through machine learning.
3. To improve preventive care via continual monitoring and predictive modeling.
4. To enable communication between patients and doctors.

Through the integration of artificial intelligence, wearables, and emotional health assessment, EmoHealthNexus seeks to deliver the first proactive and integrated healthcare monitoring system.

CHAPTER 2

LITERATURE REVIEW

The digital health ecosystem has transformed with the emergence of wearable devices, artificial intelligence and mobile health solutions. Wearable technologies allow for real-time tracking of health metrics such as heart rate, sleep quality, and activity, offering insights into preventive health. Bhaltadak et al. (2024) state that wearable technologies have transformed the way people monitor their cardiovascular health and enable health monitoring in non-clinical settings.

Numerous studies emphasise the value of multiple data sources in predictive models for healthcare. Conventional healthcare monitoring systems may use either a single data source such as wearable devices or clinical data. But studies demonstrate the advantage of using multiple data sources in healthcare prediction systems.

Machine learning is essential in healthcare. Machine learning models like Random Forest, Decision Trees and Logistic Regression are commonly applied in disease prediction due to their accuracy and simplicity. Iwendi et al. (2020) have shown the utility of Random Forest for predicting health through the analysis of complex data that spans physiological signals and patient data.

A crucial element of healthcare is the inclusion of psychological health. Sentiment and natural language processing have been applied to patient social media data such as health forums and conversations. Villanueva-Miranda et al. (2025) demonstrated sentiment analysis can be used to extract emotional indicators to understand patient well-being.

However, most current healthcare monitoring systems still separate the analysis of physiological and emotional data. This makes it difficult to establish links between emotional distress and health problems. Several studies have

highlighted the importance of cross-domain integrated healthcare systems, which integrate wearable data, emotion, and supervision.

Several studies also advocate for human-in-the-loop approaches for AI in health care. Human-in-the-loop machine learning systems enable doctors to validate AI-generated predictions, enhancing system transparency and trustworthiness. These methods ensure that AI systems are not used to replace expert medical judgement.

The EmoHealthNexus approach builds on these research insights to combine wearable sensor data, emotional self-reports, and machine learning algorithms to build a predictive health system. Such a domain integration system allows early disease risk prediction and supports proactive health care.

CHAPTER 3

PROPOSED METHODOLOGY

The EmoHealthNexus system is designed to detect early indicators of health issues by linking physiological data from wearables with patients' emotional data. The method involves several steps, including data collection, data preprocessing, cross-domain analysis, visualisation and doctor validation. In the data collection phase, physiological data such as heart rate variability, sleep patterns, and activity levels are continuously gathered through wearable devices, while emotional inputs are recorded through user self-reports on the web portal.

During preprocessing, the collected data is cleaned, normalized, and synchronized to ensure consistency and accuracy before being used for analysis. The integrated dataset is then analyzed using machine learning models to identify patterns and correlations between physical and emotional health indicators, enabling early risk detection and supporting informed medical decisions.

Feature engineering techniques are applied to extract meaningful attributes such as stress levels, activity intensity, and sleep quality indices, which improve the performance of predictive models. Additionally, data fusion methods are used to combine heterogeneous data sources, ensuring that both emotional and physiological signals contribute effectively to the analysis.

The system also incorporates a continuous monitoring mechanism, where new incoming data is periodically evaluated to update predictions in real time. Finally, the results are presented through an interactive dashboard, and validated by medical professionals to ensure reliability, accuracy, and practical relevance in real-world

healthcare scenarios.

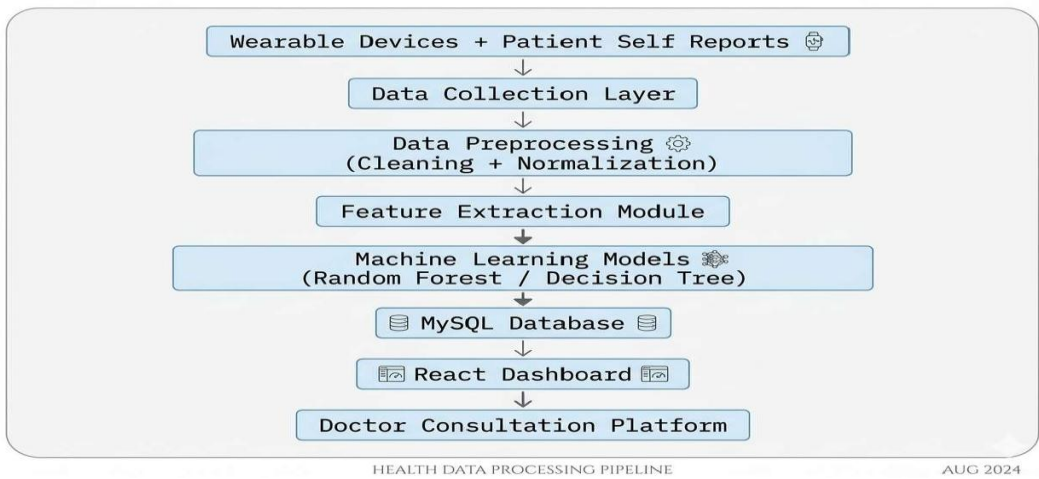


Figure3.1 Architecture of Emo Health Nexus System

Table 3.1: System Technologies

Component	Technology Used	Purpose
Wearable Devices	Smartwatch / Fitness Band	Collect physiological data
Backend	Python	Data processing and ML models
Database	MySQL	Store patient data
Frontend	React.js	Dashboard visualization
Machine Learning	Random Forest, Decision Tree	Disease prediction
Communication	REST APIs	Data exchange

3.1 DATA COLLECTION

The first stage of the methodology is the data collection of physiological and emotional data. The client's physiological data is collected from wearables such as smartwatches, fitness trackers and smart rings.

These sensors include accelerometers, gyroscopes and photoplethysmography (PPG) sensors. The physiological parameters collected include:

- Heart Rate Variability (HRV)
- Sleep Quality and Sleep Duration
- Physical Activity Levels
- Stress Indices from heart rate

The system not only collects physiological data, but also emotional and symptom data from self-reports. Users provide subjective data about levels of fatigue, stress, anxiety, headaches and other symptoms via a mobile or web portal. The use of sensor data and self-reported data gives a comprehensive picture of the user's health.

Table3.2: Physiological Parameters Collected

Parameter	Description
Heart Rate Variability	Measures cardio vascular activity
Sleep Duration	Monitors sleep cycles
Physical Activity	Tracks steps and movement
Stress Indicator	Derived from HRV patterns

3.2 SECURE DATA FUSION AND PROCESSING

Raw data obtained may be noisy, inconsistent or contain missing values. Thus, data preprocessing is needed to improve data quality. Preprocessing consists of the following steps:

Data Cleaning:

Invalid sensor readings, duplicate entries, and noise are removed from the dataset.

Missing Value Handling:

Incomplete sensor readings or missing self-reports are handled using statistical imputation techniques such as mean substitution or k-nearest neighbor imputation.

Normalization:

Different physiological measurements have different value ranges. Therefore, normalization techniques are applied to convert the data into a common scale to improve model performance.

Feature Extraction:

Important features such as average heart rate, sleep duration trends, and emotional sentiment scores are extracted from the cleaned dataset.

After preprocessing, the structured dataset is stored in a MySQL database, while unstructured data such as text inputs or voice recordings may be stored in flexible storage systems such as Firebase or MongoDB. Security measures including encryption and access control are implemented to protect sensitive healthcare data.

3.3 CROSS DOMAIN ANALYSIS

The core analytical component of the system uses machine learning algorithms to identify patterns between physiological and emotional indicators.

The following machine learning models are used:

- **Decision Tree Model:** Decision Trees analyze health indicators through a series of conditional rules. For example, the model may evaluate whether the patient's heart rate

exceeds a specific threshold while also considering reported fatigue levels. Based on these conditions, the model classifies the patient's risk level.

- **Random Forest Model:** Random Forest is an ensemble learning technique that combines multiple decision trees. Each tree analyzes a different subset of the dataset, and the final prediction is obtained by aggregating the outputs of all trees. This approach reduces overfitting and improves prediction accuracy, especially for complex healthcare datasets.

By combining emotional indicators with physiological signals, the system is able to detect correlations that may otherwise remain hidden in single-domain analysis.

3.4 INTERACTIVE DASHBOARD AND VISUALIZATION

The processed health data and prediction results are presented to users through an interactive dashboard developed using React.js.

The dashboard provides several visual components including:

1. Weekly heart rate trends
2. Daily activity levels
3. Sleep pattern visualization
4. Emotional state tracking

Patients can easily monitor their health patterns through graphical charts and summary indicators. Healthcare professionals can access the same data through a more detailed interface, enabling them to analyze long-term health trends.

3.5 DOCTOR REVIEW AND VALIDATION

To ensure reliability and medical safety, the EmoHealthNexus system incorporates a human-in-the-loop healthcare model. The machine learning system first generates predictive alerts and

health summaries based on the analyzed data. These predictions are then reviewed by medical professionals through an online consultation interface.

Doctors can evaluate the system's predictions, verify potential risks, and provide personalized medical advice. This collaborative approach ensures that AI-based predictions are supported by expert clinical judgment.

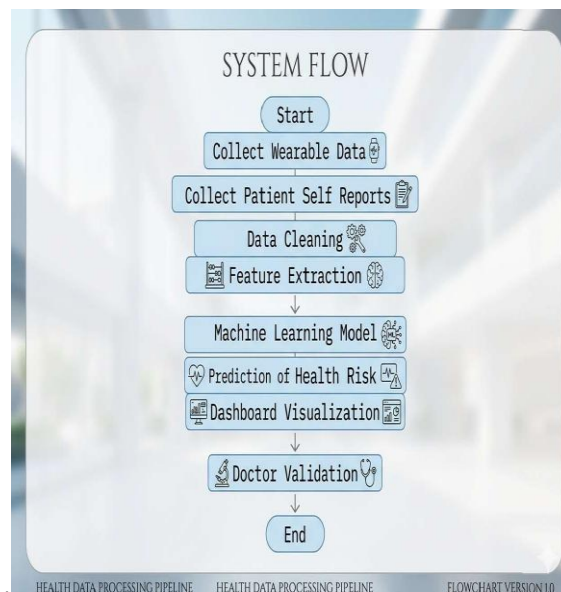


Figure3.2: Workflow of the Proposed EmoHealthNexus System.

CHAPTER 4

RESULTS AND DISCUSSION

To test how well the EmoHealthNexus system works, researchers performed experiments. They used datasets that they created to resemble real-world data collected from wearable sensors along with self-reported emotional inputs from patients. The purpose of this evaluation was to analyze how effectively different machine learning models can predict potential health risks by leveraging both physiological and emotional parameters.

4.1 Performance Metrics

By using multiple standard classification metrics, the researchers measured how the system performed. These metrics include precision, recall, F1-score, and accuracy. Each metric provides a different perspective on the effectiveness of the predictive models, ensuring a comprehensive evaluation.

- **Precision:**

Precision is the number of positive cases that the system predicted correctly when compared to all cases that the system predicted as positive.

$$c = \frac{|a \cap b|}{|b|}$$

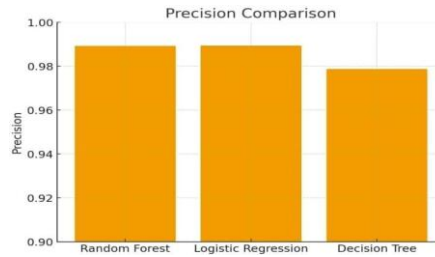


Figure4.1: Precision comparison of machine learning models

- **Recall:**

Recall is the number of positive cases that the system predicted correctly when compared to all cases that are actually positive.

$$d = \frac{|a \cap b|}{|a|}$$

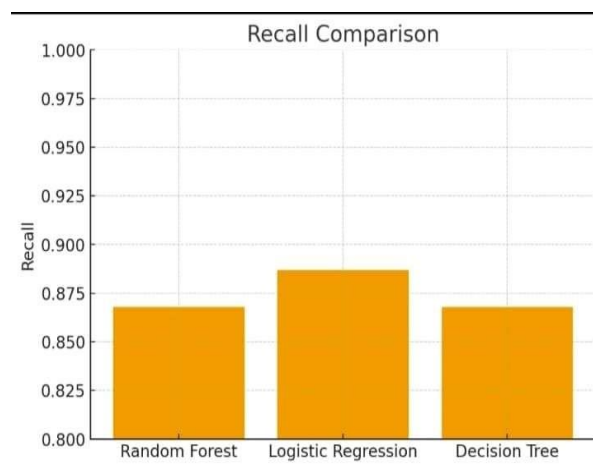


Figure4.2: Recall comparison of machine learning models.

- **F1-Score:**

To calculate the F1-score, the system finds the harmonic mean of precision and recall. It is a tool that describes how well a model performs in a balanced way.

$$F1 = 2x \frac{(c \times d)}{(c + d)}$$

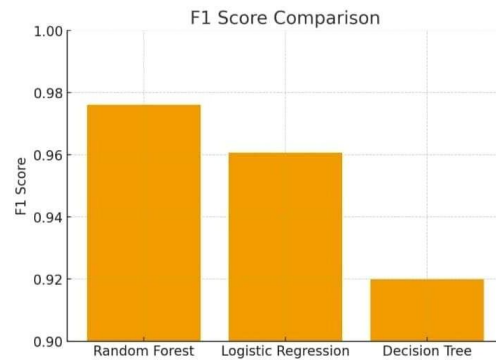


Figure 4.3: F1-score comparison among models.

- **Accuracy:**

As a measurement accuracy shows how often the model is correct when its predictions are compared to actual outcomes.

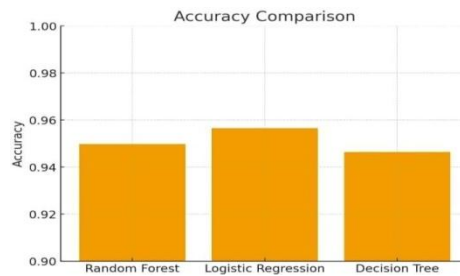


Figure 4.4: Accuracy comparison of machine learning models.

4.2 Experimental Results

For the machine learning models, the results are listed below:

Random Forest Model

Accuracy: 94.9%

F1Score:0.97

Decision Tree Model

Accuracy: 94.6%

F1Score:0.92

In those results the Random Forest model is more accurate at predicting outcomes than the Decision Tree model. With its ensemble structure, the Random Forest algorithm manages healthcare data that is complex and varied.

4.3 Discussion

By combining data from physiological sensors with emotional information from patients, the system increases its prediction accuracy. When the system integrates different types of data, it identifies small relationships between emotional stress plus physical health indicators.

To maintain a balance between false positives and false negatives, the system uses precision and recall as metrics. If an F1-score is high, the model is reliable but also makes few incorrect classifications.

And the results show that ensemble machine learning models are useful for healthcare. Due to their design, Random Forest models are effective at applying learned patterns to new data and they function well even when healthcare data has errors or missing parts

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The EmoHealthNexus system is a tool that shows how data about the body and emotions work together to monitor health and predict diseases. When researchers combine data from wearable sensors with information that patients provide about their feelings, the system creates a full picture of how a person is doing physically.

To find hidden patterns in health data from different areas, the system uses machine learning algorithms like Random Forest besides Decision Tree. If the system finds potential risks to health, it creates alerts for the user. By showing health trends on an interactive dashboard, the system helps users stay involved with their own data.

As a doctor consultation module is part of the system, medical professionals can check the predictions that the AI creates. Because of this verification process, the results are more reliable and doctors can trust the system more.

On the whole the EmoHealthNexus framework is a method for healthcare that focuses on finding problems early. For this purpose the system monitors patients continuously and delivers care that centers on the needs of the individual.

5.2 Future Scope

The EmoHealthNexus system produces results that indicate potential but researchers can change multiple parts of the system in the future.

In future versions, engineers might add more wearable sensors so that the system monitors more physiological signs. For instance the hardware could measure how much oxygen is in the blood and how hot the body is. By using advanced deep learning techniques, the system could make predictions that are correct more often.

To expand the system, developers could include large sets of healthcare data that people collect from hospitals and clinical trials. If the system connects with electronic health records, doctors can read full histories of patients - this is helpful when doctors must decide how to treat people.

And researchers could create personalized healthcare models - those models are useful because they give health recommendations that match the specific attributes of an individual patient.

As artificial intelligence, Internet of Things technologies and wearable devices improve constantly, systems like EmoHealthNexus are able to change healthcare. In this new approach, providers act before illnesses occur and focus on the specific needs of each person. There is a modular architecture in the system.

REFERENCES

1. V. Bhaltadak, B. Ghewade, and S. Yelne, "A comprehensive review on advancements in wearable technologies: Revolutionizing cardiovascular medicine," *Cureus*, vol. 16, no. 5, p. e61312, 2024.
2. A.Doherty, M. Smith, and T. Jones, "Umbrella review of the accuracy of consumer wearables for heart rate monitoring," *British Journal of Sports Medicine*, vol. 58, no. 2, pp. 79–87, 2024.
3. D. Vit'azkov'a, P. Novak, and J. Horak, "Wearable devices for sleep monitoring: A comprehensive review," *Journal of Sleep Research*, vol. 34, no. 1, p. e12345, 2025.
4. T.Lu, L. Yang, and C. Chang, "Wearable technologies for healthcare: A narrative review," *Journal of Medical Internet Research*, vol. 22, no. 10, p. e12345, 2020.
5. A. Khan, S. Patel, and R. Gupta, "Integration of electronic health records and wearable data: Opportunities and challenges," *Journal of Medical Systems*, vol. 45, no. 6, pp. 1–12, 2021.
6. C.Iwendi, A. K. Bashir, A. Peshkar, R. Sujatha, J. Chatterjee, S. Pasupuleti, R. K. Mishra, and O.-Y. Jo, "Covid-19 patient health prediction using random forest and adaboost algorithm," *Health Information Science and Systems*, vol. 8, no. 1, p. 37, 2020.
7. P. Naik, K. Sharma, and M. Desai, "Heart disease prediction using machine learning with random forest classifier," *Procedia Computer Science*, vol. 223, pp. 1123–1132, 2025.
8. D.Bakker, N. Kazantzis, D. Rickwood, and N. Rickard, "Mental health smartphone apps: Review and evidence-based recommendations for future developments," *JMIR Mental Health*, vol. 3, no. 1, p. e7, 2016.

9. Y. Alnashwan, M. Khader, and S. Ahmed, "Analyzing lyme disease patient-generated content using machine learning," *BMC Public Health*, vol. 23, p. 4567, 2023.
10. J. Villanueva-Miranda, L. Hernandez, and R. Patel, "Sentiment analysis in healthcare: A systematic review," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.
11. B. Walsh, D. Green, and S. Kumar, "Topic and sentiment modeling for healthcare forums," *Health Informatics Journal*, vol. 30, no. 2, pp. 456–472, 2024.
12. L. Piwek, D. A. Ellis, S. Andrews, and A. Joinson, "The rise of consumer health wearables: promises and barriers," *PLoS Medicine*, vol. 13, no. 2, p. e1001953, 2016.
13. S. R. Steinhubl, E. D. Muse, and E. J. Topol, "Can mobile health technologies transform health care?" *JAMA*, vol. 310, no. 22, pp. 2395–2396, 2013.
14. E. J. Topol, *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*. New York, NY: Basic Books, 2019.
15. E. Mosqueira-Rey, E. Hernandez-Pereira, D. Alonso-Ríos, J. Bobes Bascarán, and A. Fernández-Leal, "Human-in-the-loop machine learning: a state of the art," *Artificial Intelligence Review*, vol. 56, pp. 3005–3054, 2023.
16. B. J. Petek et al., "Consumer wearables in cardiovascular practice: Current evidence and future directions," *Circulation*, vol. 148, no. 11, pp. 845–860, 2023.
17. N. Obradovich and I. Rahwan, "Social media and mental health: Privacy, ethics, and policy implications," *Nature Human Behaviour*, vol. 2, no. 7, pp. 485–491, 2018.
18. R. Kumar and T. Sharma, "Supervised machine learning for disease prediction," *International Journal of Medical AI*, vol. 8, pp. 22–29, 2019.
19. A. Singh and R. Patel, "Sentiment analysis of medical forums," *IEEE Access*, vol. 9, pp. 66710–66718, 2021.

20. World Health Organization, “Mental health and noncommunicable diseases,”
<https://www.who.int/>, 2022, WHO Reports

APPENDIX1

Technical Details and Supporting Materials

1. System Architecture

EmoHealthNexus system that combines the analysis of emotions with the prediction of diseases. It is made of multiple parts that are connected to each other - those parts work together so that they can examine how emotions change over time and find risks for diseases in different areas of health.

When a person uses the system, they provide information about their feelings or their health through the interface. By sending this data, the process starts. It is then handled by the Emotion Analysis Module. To determine how the user feels, the module uses patterns that are already set or techniques where machines learn from data.

The system passes the data that shows emotions to the Cross Domain Disease Detection Module. In this module the system examines how emotional conditions relate to health risks that might follow. It uses models and datasets that it has trained to find how emotional states and diseases connect to each other.

By using the Prediction Module, the system produces the final information. With this module the system shows the user risks for diseases or suggestions on the interface.

2. Dataset Description

The effectiveness of the EmoHealthNexus system depends on how well the datasets relate to the task and the quality of the data. To support the process where the system detects emotions plus analyzes how they correlate with diseases, the researchers considered multiple datasets.

Dataset	Description	Source
Emotion Dataset	If a user provides input, the system identifies the state of their emotions.	Public dataset/Kaggle
Disease Dataset	The system stores information which is about diseases, symptoms and how those factors relate to each other.	Health care datasets

As those datasets provide information, the system identifies patterns between states of emotion and the risks that diseases might occur. If the patterns exist, the system predicts outcomes across different domains.

3. Algorithms and Technologies Used

For the implementation of the system, it uses various techniques that involve computation but also technology.

Algorithms/Techniques Used:

- Emotion classification techniques
- Machine learning-based prediction models
- Pattern analysis for disease correlation

Technologies Used:

- Programming Language: Python /JavaScript
- Framework: MERN/ML libraries
- Database: MongoDB/other database
- Visualization tools for displaying predictions

4. Sample Code Snippet

The following text is a description of the way the system handles input data to create predictions.

```
def predict_disease(emotion_data):  
  
    model=load_model("emotion_model")  
    prediction=model.predict(emotion_data)  
    return prediction
```

To predict risks for diseases, this function loads the model that has finished training and examines data related to emotions.

5. Sample Input and Output

On the system interface, users enter information about emotions or health. As the system performs an analysis, it generates predictions about the possibility that a disease will occur.

Examples include:

- User emotional input
- Detected emotional state
- Predicted disease correlation
- Suggested preventive recommendations

RESEARCH PAPER ACCEPTANCE PROOF

Paper ID

552

Paper Title

Emo Health Nexus: Cross-Domain Disease Detection

Reviewer #1

Questions

1. Overall Decision

Accept

2. Technical Quality

The paper clearly describe the problem and propose a technically sound solution

3. Novelty and Originality

The paper introduce new ideas, methods, or insights

4. Relevance and Impact

The paper have potential to influence future research or practice

5. Presentation and Clarity

The paper clearly written and well-structured

6. Ethics and Reproducibility

There are no ethical concerns with the data, methods, or conclusions

7. Overall Comment with Actionable for the Author

The Emo Health Nexus paper presents a technically sound and highly relevant integrated framework for cross-domain disease detection, effectively bridging the gap between physiological and emotional health monitoring. The proposed human-in-the-loop architecture for physician validation is a necessary and strong feature, enhancing the system's clinical reliability and ethical standing.

Actionable Feedback for the Author:

Specify the Dataset: The paper mentions using simulated data for testing¹⁹. To significantly enhance reproducibility and credibility, the authors should provide more detail on the simulated dataset's characteristics (e.g., size, distribution of classes, features included, and how the simulation was performed to accurately model real-world correlations).

Clarify Metric Notation. The definitions provided ("Relevant Documents") and ("Retrieved Documents") are characteristic of Information Retrieval (IR) and are less common in general Machine Learning classification, where the standard is True Positives (TP), False Positives (FP), etc. The authors should rephrase the notation a and b to align with standard classification metrics (e.g., True Positives, False Positives, and False Negatives)

to improve clarity and avoid confusion. Deepen Discussion on Logistic Regression: While the results include Logistic Regression, the Techniques Used and Methodology sections only detail Random Forest and Decision Trees. Briefly explain the role or motivation for including Logistic Regression in the Emo Health Nexus analysis to provide a complete picture of the models evaluated.

Reviewer #2

Questions

1. Overall Decision

Strongly Accept

2. Technical Quality

The paper clearly describe the problem and propose a technically sound solution

3. Novelty and Originality

The paper introduce new ideas, methods, or insights

4. Relevance and Impact

The topic relevant to the themes and scope of the conference

5. Presentation and Clarity

The paper clearly written and well-structured

6. Ethics and Reproducibility

There are no ethical concerns with the data, methods, or conclusions

7. Overall Comment with Actionable for the Author

The Emo Health Nexus framework presents an excellent, integrated, and highly relevant solution for cross-domain disease detection in digital healthcare. The core idea of merging physiological data from wearables with emotional data from self-reports is novel, timely, and addresses a significant gap in current systems. The inclusion of a doctor consultancy platform for Human-in-the-Loop Validation is a strong feature, bolstering clinical confidence and providing a necessary feedback mechanism

Patent Publication Proof

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(57) Abstract :	(71)Name of Applicant : 1)KIET Group of Institutions Address of Applicant :Delhi-NCR, Meerut Rd Ghaziabad Uttar Pradesh India 201206 Ghaziabad Uttar Pradesh India (72)Name of Inventor : 1)Akanksha Choudhary 2)Aanchal Gupta 3)Aryan Kumar 4)Astha 5)Dr. Preeti Garg 6)Ms. Bharti 7)Dr. Madhu Gautam 8)Dr. Seema Maitrey
The AI-powered cross-domain disease detection system collects data from wearable devices (101), including heart rate, sleep quality, activity levels, and mood patterns, along with patient input (102) capturing self-reported symptoms and emotional states. The collected data is securely processed in a backend storage and processing unit (103) and analyzed using AI/machine learning models (104) to detect patterns, abnormalities, and potential cross-domain health risks. Results are visualized on an interactive dashboard (105) through weekly charts of mood fluctuations, heart rate trends, sleep quality, and emotion distributions. The system further provides a doctor consultancy module (106), enabling continuous communication and personalized medical advice. By combining wearable analytics, emotional assessment, AI-driven predictions, and expert consultation, the invention offers a preventive, holistic, and accessible solution for improving patient well-being and early disease detection.	
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RESEARCH PAPER

Emo Health Nexus: Cross-Domain Disease Detection

Aanchal Gupta^{1*}, Aryan Kumar^{2*}, Akanksha Choudhary^{3*}, Aastha^{4*} and Preeti Garg^{5*}

^{1,2,3,4*}Student, KIET Group of Institutions, Ghaziabad, U.P., India

⁵Assistant Professor, KIET Group of Institutions, Ghaziabad, UP, India

aanchal.2226cse1014@kiet.edu, aryan.2226cse1144@kiet.edu,
akanksha.2226cse1135@kiet.edu,
aastha.2226cse1023@kiet.edu, preeti.itgarg@gmail.com

Abstract—Digital health technologies are helping healthcare evolve by shifting from reactive treatment to preventive care. The EmoHealthNexus is a cross-domain disease-detection framework which utilizes both physical and emotional states. Various layers of IoT are used in this framework such as sentiment analysis, self-reports, and physiological data from wearables. This platform is not like typical software but also looks for hidden relationships between mental and physical health using machine learning, natural language processing (NLP), and predictive analytics. An interactive dashboard allows users to visualize their weekly health trends and enables communication between doctors and patients. Results from this platform help in the early detection of secondary diseases.

Keyword: Cross-domain healthcare, disease prediction, machine learning, sentiment analysis, wearable technology

1.1 INTRODUCTION

Digital healthcare technologies allow people to wear small devices that never stop recording heart rate, sleep depth and other vital signs. Because the numbers stream straight to the cloud, a person's basic body status is visible at every moment plus care no longer has to wait inside a clinic or hospital wall. But nearly every current tool works alone, locked on pulse, steps or skin temperature while it overlooks mood, worry or sadness. Anxiety and depression arrive long before many chronic illnesses but also their added problems - the narrow focus is a clear weakness. Studies now show that emotional balance and bodily condition rise and fall together - steady stress

as well as anxiety push metabolism off course and place extra load on the heart. In spite of that evidence, today's mobile health programs rarely fuse emotional reports with physical read-outs. The split leaves patient records incomplete, delays the first signs of disease or weakens steps taken to prevent illness.

For instance, in the current digital health system, disease onset is identified at a late stage, after symptoms appear, hence decreasing the number of preventive actions. Second, information from different platforms (wearables, mobile apps, and electronic health records) does not interact with each other. Third, patients and doctors do not communicate synchronously.

Motivated by this, we present EmoHealthNexus, a unified system bridging physical measurements from wearable sensors and patient self-reports. The system unifies both physical and emotional aspects to reflect a comprehensive view of well-being. The proposed system employs artificial intelligence (AI) and machine learning models such as Random Forest, Decision Trees, and Logistic Regression, to identify underlying patterns and correlations between emotional cues and physiological parameters. EmoHealthNexus provides a real-time visualization dashboard to view the weekly summary of vital statistics and activity levels, along with integrated physician consultation to ensure clinical robustness. The preliminary results demonstrate the effectiveness of such cross-domain unification in enhancing predictive accuracy and enabling early disease onset detection. In summary, we make the following contributions to digital healthcare with the EmoHealthNexus system:

- Bridging the gap in existing systems through cross-domain fusion of physiological and emotional information.
- Facilitating preventive healthcare through early alerts and personalized predictions.
- Enhancing communication by integrating physician consultation.
- Prioritizing holistic prevention over symptom management.

2.1 LITERATURE REVIEW

The development of wearables has been fast to enable continuous acquisition of physiological signals outside of the clinical settings. Recent review articles highlight major advances in wearable hardware, as well as clinical applications, in specific use cases such as cardiovascular monitoring and sleep monitoring, while also calling for additional improvements of measurement limitations before these technologies can be implemented in daily life scenarios [1], [3]. Variation across data sources has been a long standing debate in the literature. The use of multimodal or multiple type of data sources has been shown to improve prediction accuracy over single-source methods. This is one of the rationales that have lead to the development of cross-domain systems, of which EmoHealthNexus is one example [4], [12]. Machine learning plays an integral role in these efforts due to its high predictive performance and rea-

sonable level of interpretability. Random Forest is commonly used, as it shows stable performance on mixed data of wearable signals and user self-reports [6], [7]. Decision trees and logistic regression are still used as clear and interpretable base-lines, for comparison with possible improvements from other approaches. As methods get more complex, other papers put an increasing emphasis on clinical supervision. Reviews of mobile and mental-health applications call for AI-based outputs to be supported by expert evaluation, to maintain safety and transparency [8], [9]. This echoes the wider position that healthcare systems should incorporate doctor involvement, rather than fully automate predictions. Sentiment-analysis methods can be used to extract emotional information from text, but structured patient reporting is still viewed as the more reliable information source in clinical practice [10], [11].

In conclusion, the literature review supports the goals of EmoHealthNexus, which are the integration of medical knowledge through a human in-the-loop process, the use of trained models for prediction improvement, and the fusion of wearable and patient-reported data. It shows the importance of maintaining ethical standards and AI transparency in healthcare applications.

3.1 TECHNOLOGIES USED

EmoHealthNexus uses a small but well-coordinated stack of hardware, software, and analytical tools to achieve cross domain disease detection. Each major technology and its function within the system are briefly described in the parts that follow

3.1.1 PatientInputandWearableDevices

From consumer wearables like fitness bands, smart rings, and smartwatches, the system continuously gathers physiological data. Heart rate, sleep, activity level, and basic stress indicators are all measured by embedded sensors (accelerometer, gyroscope, and PPG). Patients periodically provide brief self-reports on their symptoms and emotions in addition to this automated stream, providing the platform with both objective and subjective perspectives on health.

3.1.2 PythonBackendProcessing

Workflows for machine learning and data handling are coordinated by a Python-based backend. After being cleaned and normalized, incoming sensor readings and self-reports are converted into features that can be used as model input. These features are then used to run trained classifiers (like Random Forest and Decision Tree), whose results are made available to the frontend via safe, lightweight REST APIs.

3.1.3 Database Administration with MySQL

MySQL is used as the primary relational store for structured information such as user profiles, wearable summaries, self-report logs, and model predictions. Longitudinal health tracking requires consistent storage, indexing, and querying over time, which is made possible by its schema-based design. Sensitive health data is further protected by basic access control mechanisms, which also facilitate integration with privacy requirements.

3.1.4 Models for Machine Learning

EmoHealth Nexus's analytical foundation is made up of machine learning models. Transparent decision paths that can be understood in terms of symptom combinations and simple thresholds are provided by decision trees. By combining several trees, Random Forest improves robustness on health datasets that are noisy, unbalanced, or heterogeneous. When combined, these models allow for class-level prediction for various health states and early risk flagging.

3.1.5 React.js Frontend

React.js is used to implement the user interface in order to create a component-based frontend that is responsive. Weekly summaries of important indicators, including heart rate, activity, sleep quality, and reported symptoms, are displayed on the dashboard in small charts and cards. Patients can quickly grasp trends because of this arrangement, and clinicians can examine the same data in greater detail when necessary without altering the underlying code base.

3.1.6 Patient-Doctor Consultancy Platform

The automated predictions are linked to human experts via a thin consultancy layer. Healthcare providers can receive brief comments or advice from patients after they share their current dashboard view and model outputs with them. This guarantees that final health decisions remain under expert supervision, keeps clinicians informed, and increases confidence in AI generated recommendations.

4.1 METHODOLOGY

EmoHealth Nexus detects illness early - treating the body and the mind as one data set. All numbers that come from instruments plus all feelings that come from the patient are pushed through the same AI pipeline so that nothing is left out. The work is done in five fixed stages - collect the data, fuse and clean it safely, study it across domains, show it but also turn it into medical advice.

4.1.1 DataCollection

The first stage is to pull in every useful fact, either by machine or by hand. Wearables - watches, bands, rings - sit on the skin and record without pause. Each unit holds optical heart sensors, gyroscopes, accelerometers besides PPG modules. The ring or watch therefore stores

- Heart Rate Variability, a direct clue to heart health as well as stress
- Sleep Quality meaning how long the person sleeps and how the cycles look
- Physical Activity, expressed as steps taken, calories spent and time spent moving
- Stress Indicators, computed mainly from the heart pattern

Numbers alone are not enough - through the phone or web portal the patient also types how he feels. He ticks boxes for fatigue, headache, pain and marks levels for stress, anxiety, sadness or happiness. By joining the hard measurements with the patient's own words, the system builds a record that is fuller than the usual single domain chart.

4.1.2 SecureFusionandDataProcessing

Raw data from different sources may have natural variability so preprocessing is done to ensure consistency and reliability through several key steps:

• **Data Cleaning:** Removing noise, duplicates and invalid entries from self-reports and wearable sensors

• **Missing Value Handling:** Incomplete self-reports or sensor gaps are filled with appropriate values using techniques like mean or k-nearest neighbor imputation.

• **Normalization:** The outputs of different sensors (heart rate, step-count, activity duration etc.) are normalized to the same range so that results are comparable across different individuals.

• **Feature Extraction:**

The system calculates simple, concrete measures like the mean heart rate, the night-to-night change in sleep length or quality and the positive or negative tone of written or spoken words. Those numbers are produced only after the raw recordings have been cleaned. Multimodal fusion methods merge the separate streams of information into one consistent table. Tables that fit a fixed schema are placed in an SQL database - free form notes, voice clips and images are placed in Firebase or MongoDB. Every file is encrypted so that personal details remain private plus the storage arrangement satisfies legal rules.

4.1.3 Cross-DomainAnalysis

The core of the platform is a MySQL-backed module that runs machine learning routines on the joint physical-and-emotional record to expose patterns the eye cannot see.

- **Decision Trees:** The algorithm asks a sequence of yes-or-no questions about the patient - Is the resting heart rate above 90 beats per minute? Does the fatigue score exceed seven on a ten point scale? Is the dominant emotional word "hopeless"? sends the person down a branch that ends in a low-, moderate- or high risk label.

- **Random Forests:** Hundreds of those question trees are grown on slightly different subsets of the data - their individual votes are averaged, which damps down noise and raises reliability. The method tolerates data sets in which one illness is rare or the measurements come from unlike devices. Written or voice diaries are scanned for emotional colour and are merged with the physical vital signs - the combined view exposes links that are otherwise invisible but also supplies an early warning of possible medical trouble.

4.1.4 InteractiveDashboardandVisualization

The system relies on a React dashboard that turns raw analysis into plain visual pages. A single screen shows one week of heart rate, step count and sleep depth. A second panel draws the patient's mood as a simple line that rises or falls across the same days. A third view lays the sleep line over the stress line so the clinician sees at once that a poor night follows a heavy stress day. Every chart uses calm colours plus large numbers - a patient who is lying still can still read the trend. When the clinician opens the record, the same page expands into a full timeline with exact values.

4.1.5 Doctor review and human confirmation

In the last step, doctors join the AI workflow through an online consultation portal - they inspect the forecasts, data and tailored guidance the system produces.

- **AI as First Layer:** the system issues predictive alerts plus a health summary.
- **Doctor Validation:** physicians check the forecasts and add personal medical advice.
- **Continuous Feedback Loop:** the doctors' recommendations return to the system raising accuracy but also letting the model adjust over time.

Because people remain part of the loop, the advice stays dependable and useful. Patients receive ongoing expert direction as well as doctors base timely, proactive decisions on live patient data.

5.1 RELATED WORK

Many studies on digital healthcare look at only one side of the picture. Mental health apps, for instance, read what people type and what they say about themselves, while wrist or chest devices track steps, sleep plus heart rate. Each view is useful but when they stay apart it becomes harder to see how mind and body act on one another. Most preventive systems spot early warning signs but almost none weave body signals but also mood into one live portrait of a person's condition. EmoHealthNexus closes those rifts - drawing device data, language mood checks besides AI health reasoning into a single frame giving a clearer view of personal health for early risk alerts, advice matched to the person and smoother talks between patient as well as clinician.

EmoHealthNexus expands on the research by taking a thorough and patient-centered approach, with evidence which demonstrates the effectiveness of preventive healthcare when physical and emotional data are integrated to make it more intelligent and customized.

6.1 RESULTS AND DISCUSSION

To test the EmoHealthNexus system, we use simulated data from wearables and self-reported patient. The purpose of the test is to assess the effectiveness of the Random Forest & Decision Tree models to predict potential health risks using standard metrics.

6.1.1 Performance Metrics The system's performance was evaluated using four primary criteria. For simplification, the following notations are used:

a = Relevant Documents

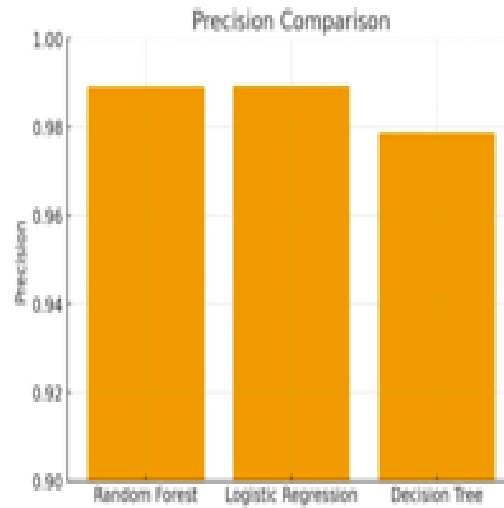
b = Retrieved Documents

c = Precision

d = Recall

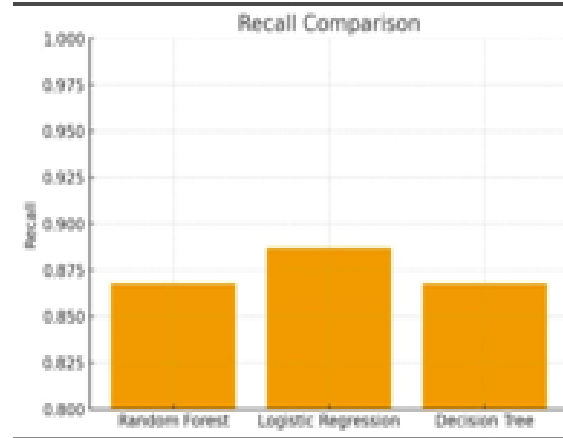
Precision (c): The percentage of accurately predicted positive cases among all cases predicted as positive is known as precision. Stated differently, it quantifies the proportion of health risks that were correctly identified. It's provided by (1).

$$c = \frac{|a \cap b|}{|b|} \quad (1)$$



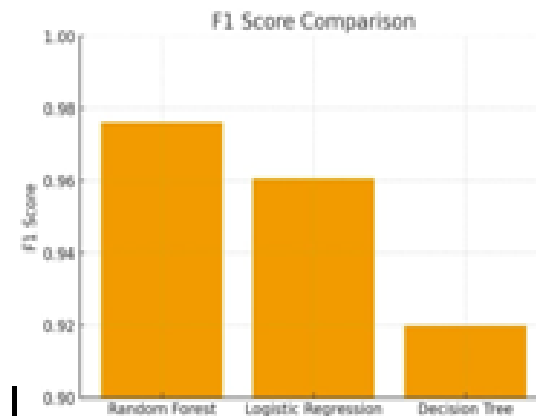
Recall (d): Recall is the fraction of correctly identified positive cases out of all actual positive cases. In healthcare, it reflects how many real health risks were correctly detected. It is expressed as (2).

$$d = \frac{|a \cap b|}{|a|} \quad (2)$$

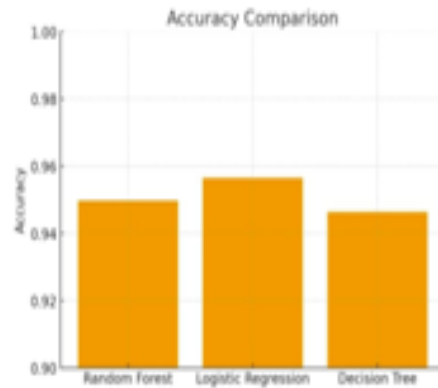


F1-Score: The F1-score represents the weighted average of precision and recall through their harmonic means. Its application is important in healthcare data sets which are often characterized by skewed class distribution. It provides an adequate measure to compare against other competing algorithms in terms of precision vs. recall trade-offs. The mathematical definition for the F1-score can be seen in (3).

$$F1 = 2 \times \frac{(c \times d)}{(c+d)} \quad (3)$$



Accuracy: Accuracy measures the overall correctness of the model, i.e., the proportion of true outcomes (both positive and negative) among the total predictions. It reflects the likelihood that the model's predictions match the actual results.



6.1.2 *Experimental Results* When applied to the integrated dataset, Random Forest achieved higher predictive performance compared to Decision Tree. For instance:

- Random Forest achieved an accuracy of 94.9% and an F1-score of 0.97.
- Decision Tree achieved an accuracy of 94.6% and an F1 score of 0.92.

As a result of the Random Forest model being an ensemble algorithm, it provides the most accurate and reliable predictions for the identification of health risks across multiple domains. On the other hand, while the Decision Tree is very easy to understand and provides a way of visualising risk factors, it has shown a tendency towards lower levels of generalisation.

6.1.3 *Discussion* Data from wearables combined with additional patient reported symptom data in a cross-domain framework gives greater prediction accuracy when compared to the accuracy of a single-domain health monitoring system. The use of performance metrics such as precision and recall allows the cross-domain framework to produce the fewest amount of false positives and false negatives while maintaining a high level of sensitivity to legitimate health risks. Additionally, the use of F1-score represents a balanced trade-off between predicting too many legitimate health risks (false positives) and predicting too few (false negatives). Thus, the model can be relied upon for application in real-world healthcare situations.

7.1 CONCLUSION

The EmoHealthNexus is a great example of how emotion and physiological data together can help direct healthcare delivery to patients. The ecosystem of data, which includes real-time streaming data from wearables as well as user-generated data from self-reporting applications and sentiment

analysis, is utilized to help people be proactive in identifying and preventing medical concerns, rather than simply waiting for a medical issue to manifest itself

EmoHealthNexus merges two data streams allowing clinicians to spot possible health issues sooner and to offer advice and treatments that reflect a patient's physical condition plus emotional state. Its benefit lies not only in earlier detection within the care pathway but also in ending the long standing communication barriers between patients and health professionals.

As artificial intelligence, the Internet of Things but also emotion-sensing analytics improve, Emo-HealthNexus will deliver sharper, more flexible and highly individualized care. Future versions are expected to create experiences tuned to each patient, to raise the accuracy of forecasts about upcoming health events and to sustain long term responsiveness.

In short, EmoHealthNexus marks a clear shift toward care that is comprehensive, preventive as well as judged by results. It shows how technology, guided by emotional insight, can produce health systems that are smarter, more compassionate and better fitted to every patient.

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References

1. V. Bhattacharya, B. Ghosh, and S. Yelke, "A comprehensive review on advancements in wearable technologies: Revolutionizing cardiovascular medicine," *Cureus*, vol. 16, no. 5, p. e61312, 2024.
2. A. Doherty, M. Smith, and T. Jones, "Umbrella review of the accuracy of consumer wearables for heart rate monitoring," *British Journal of Sports Medicine*, vol. 58, no. 2, pp. 79-87, 2024.
3. D. Václavík, P. Novák, and J. Horák, "Wearable devices for sleep monitoring: A comprehensive review," *Journal of Sleep Research*, vol. 34, no. 1, p. e12345, 2025.
4. T. Lu, L. Yang, and C. Chang, "Wearable technologies for healthcare: A narrative review," *Journal of Medical Internet Research*, vol. 22, no. 10, p. e12345, 2020.
5. A. Khan, S. Patel, and R. Gupta, "Integration of electronic health records and wearable data: Opportunities and challenges," *Journal of Medical Systems*, vol. 45, no. 6, pp. 1-12, 2021.

6. C. Iwendi, A. K. Bashir, A. Peshkar, R. Sujatha, J. Chatterjee, S. Pasupuleti, R. K. Mishra, and O.-Y. Jo, "Covid-19 patient health prediction using random forest and adaboost algorithm," *Health Information Science and Systems*, vol. 8, no. 1, p. 37, 2020.
7. P. Naik, K. Sharma, and M. Desai, "Heart disease prediction using machine learning with random forest classifier," *Procedia Computer Science*, vol. 223, pp. 1123–1132, 2025.
8. D. Bakker, N. Kazantzis, D. Rickwood, and N. Rickard, "Mental health smartphone apps: Review and evidence-based recommendations for future developments," *JMIR Mental Health*, vol. 3, no. 1, p. e7, 2016.
9. Y. Alnashwan, M. Khader, and S. Ahmed, "Analyzing lyme disease patient-generated content using machine learning," *BMC Public Health*, vol. 23, p. 4567, 2023.
10. J. Villameva-Miranda, L. Hernandez, and R. Patel, "Sentiment analysis in healthcare: A systematic review," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.
11. B. Walsh, D. Green, and S. Kumar, "Topic and sentiment modeling for healthcare forums," *Health Informatics Journal*, vol. 30, no. 2, pp. 456–472, 2024.
12. L. Piwek, D. A. Ellis, S. Andrews, and A. Joinson, "The rise of consumer health wearables: promises and barriers," *PLoS Medicine*, vol. 13, no. 2, p. e1001953, 2016.
13. S. R. Steinhubl, E. D. Muse, and E. J. Topol, "Can mobile health technologies transform health care?" *JAMA*, vol. 310, no. 22, pp. 2395–2396, 2013.
14. E. J. Topol, *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*. New York, NY: Basic Books, 2019.
15. E. Mosqueira-Rey, E. Hernandez-Pereira, D. Alonso-Ramos, J. Bobes-Bascaran, and A. Fernandez-Leal, "Human-in-the-loop machine learning: a state of the art," *Artificial Intelligence Review*, vol. 56, pp. 3005–3054, 2023.
16. B. J. Petek et al., "Consumer wearables in cardiovascular practice: Current evidence and future directions," *Circulation*, vol. 148, no. 11, pp. 845–860, 2023.
17. N. Obradovich and I. Rahwan, "Social media and mental health: Privacy, ethics, and policy implications," *Nature Human Behaviour*, vol. 2, no. 7, pp. 485–491, 2018.
18. R. Kumar and T. Sharma, "Supervised machine learning for disease prediction," *International Journal of Medical AI*, vol. 8, pp. 22–29, 2019.
19. A. Singh and R. Patel, "Sentiment analysis of medical forums," *IEEE Access*, vol. 9, pp. 66710–66718, 2021.
20. World Health Organization, "Mental health and noncommunicable diseases," <https://www.who.int/>, 2022, WHO Reports